

LoRA-Integrated Faithfulness-Aware Scientific Summarization Using Large Language Models

Mushfiquur Rahman Rafin
Computer Science
University of Missouri-Kansas City Mo, USA
mrafin@umkc.edu

ABSTRACT

Scientific document summarizing is a tough issue in natural language processing, aiming to reduce the amount of technical content while keeping factual consistency and semantic meaning. Large language models (LLMs) have been shown to perform quite well on the task of summarization. However, full-parameter fine-tuning of these models is costly and time-consuming. This study investigates the application of parameter-efficient fine-tuning technique, Low-Rank Adaptation (LoRA) for scientific summarization using the FLAN-T5-small model. This develop and analyze a comprehensive end-to-end summarization pipeline, comparing a fully trainable base model with a LoRA modified model under the same conditions. Experimental results show that the summarization quality of the LoRA-adapted model is greatly improved while just 2.25% model parameters are updated. The LoRA model decreases the validation loss from 1.4907 to 0.4016 and improves the ROUGE-L from 0.3244 to 0.6687. These results show that parameter-efficient fine-tuning can reach competitive summarizing performance with a large reduction in computational cost..

KEYWORDS

Scientific summarization, large language models, FLAN-T5, LoRA, parameter-efficient fine-tuning, ROUGE evaluation, natural language processing

1 INTRODUCTION

The fast rise of scientific publications presents a tremendous challenge to researchers who must process and understand large volumes of technical information. Automated summarizing methods have proven increasingly essential in helping to reduce reading time and to make information more accessible. Scientific summary focuses on producing succinct and useful summaries that retain the important results, techniques, and conclusions of scientific papers.

Recent improvements in transformer-based large language models have greatly enhanced text production tasks including translation, summarization, and question answering. Models like T5 and FLAN-T5 show powerful sequence-to-sequence creation ability via large-scale pre-training and instruction tuning. Despite these gains, applying big language models to domain-specific tasks

generally involves full-parameter fine-tuning, which is computationally demanding and memory hungry.

Parameter-efficient fine-tuning strategies aim to reduce the processing needs by updating a very small fraction of model parameters. One of the most successful methods is Low-Rank Adaptation (LoRA), where trainable low-rank matrices are inserted into the attention layers of transformers, while keeping the original pre-trained weights frozen.

This research studies whether LoRA can increase scientific summarization quality with a drastic reduction in the number of trainable parameters. The project develops an end-to-end summarization pipeline using FLAN-T5-small, and compares a fully trainable base model with a LoRA-adapted model using quantitative evaluation measures.

The key contributions of this paper are:

- FLAN-T5-small to build an end-to-end scientific summarization pipeline.
- Parameter-efficient fine-tuning with LoRA integration.
- Full-parameter vs. LoRA-adapted models quantitative comparison.
- Trade-off analysis: computational efficiency versus summarization quality

2. RELATED WORK

Vaswani et al. [1] presented Transformer architectures that replaced recurrent computing with self-attention techniques, revolutionizing natural language processing. Later, the T5 framework consolidated NLP tasks into a text-to-text paradigm [2], which has shown high performance on summarization and generation .

FLAN-T5 [3] is an instruction tuning-based extension of T5 that shows better zero-shot and few-shot learning performance. Instruction-tuned models are particularly strong in generation tasks like structured prompt generation and summarization generation.

Scientific summarizing continues to be a challenging task since the summarized content needs to be factually consistent and compact with dense technical knowledge. Traditional evaluation

metrics like ROUGE are based on lexical overlap and may not necessarily measure factual correctness [4].

Parameter-efficient fine-tuning approaches like adapter tuning, prefix tuning and LoRA have become popular alternatives to full fine-tuning. LoRA [5] injects trainable low-rank matrices into transformer attention layers and freezes original parameters, drastically cutting down memory and processing needs.

Recent work shows that LoRA can match the performance of full fine-tuning on downstream NLP tasks, while updating a tiny percentage of model parameters [6]. However, there has been little research on LoRA for scientific summarizing tasks in low-resource contexts.

3. DATASET AND PREPROCESSING

This study uses a dataset consisting of scientific type text and their related summaries in JSONL format. In each dataset sample there are two fields: paper_text for the input scientific content and summary for the reference summary. The dataset was initially developed with a modest collection of 20 samples and later expanded to a more robust 100-sample synthetic scientific summarization corpus. The larger dataset consists of subjects on scientific summarization, hallucination detection, retrieval-augmented generation, parameter-efficient fine-tuning, and low-resource natural language processing.

To enhance realism and diversity, the synthetic dataset contains organized explanations of study objectives, techniques, findings and limits. While synthetic datasets may not completely reflect real world scientific corpora, they offer a controlled environment to evaluate adaption strategies and summarization behavior. The dataset was randomly shuffled with a fixed random seed, then split into train, validation and test sets, in the ratio of 70/15/15. Experiments were reproducible due the usage of fixed seeds. Preprocessing included whitespace normalization, elimination of formatting artifacts, trimming to set sequence lengths, and tokenization using SentencePiece. To keep the training calculation efficient, input sequences were limited to 512 tokens and target summaries to 96 tokens.

Tokenization was performed with the pretrained SentencePiece tokenizer of FLAN-T5. The tokenizer has a vocabulary of about 32,100 subword tokens. Subword tokenization alleviates rare scientific terms and alleviates sequence fragmentation.

4. METHODOLOGY

FLAN-T5-small is the base model for this work. It is an encoder-decoder transformer model with roughly 76.9 million parameters. The model does sequence to sequence generation with self attention, encoder-decoder cross attention and autoregressive decoding. We chose FLAN-T5-small due to its good summarizing performance and its computational feasibility in a Google Colab setting.

LoRA was integrated into transformer attention layers to improve adaptation efficiency. LoRA implements a trainable low-rank matrices into query and value projection modules, while leaving the original pretrained model weights frozen. This allows the model to

specialize for downstream tasks without updating the entire set of parameters.

The LoRA configuration used in this study uses a rank value of 16, an alpha scaling factor of 32, and a dropout rate of 0.05. In particular, adaptation was used to the query and value projection matrices of the transformer attention mechanism. In this setup, the trainable parameters in the LoRA-adapted model were 1,769,472, or about 2.25% of the total parameters, compared to 76,961,152 for the fully trainable base model.

Cross-entropy loss optimization was utilized to train the model using the AdamW optimizer. It trained the base model for three epochs with a learning rate of $2e-5$ and the LoRA-adapted model for 15 epochs with a learning rate of $3e-4$. All studies used a batch size of 4. For generation, this employed beam search decoding with three beams to increase the quality of the summaries and reduce repetitive outputs. It also set a minimum output length of twenty-five tokens, repetition penalties, and no-repeat n-gram constraints.

5. EXPERIMENTAL RESULTS

The models were evaluated using ROUGE-1, ROUGE-2, and ROUGE-L metrics. ROUGE-1 is the unigram overlap, ROUGE-2 is the bigram overlap and ROUGE-L is the longest common subsequence similarity between the generated and reference summaries.

Table 1: Base vs LoRA Comparison

Model	Trainable %	Validation Loss	ROUGE -1	ROUGE -2	ROUGE -L
Base FLAN-T5-small	100.0%	1.4907	0.3880	0.2361	0.3244
LoRA FLAN-T5-small	2.2475%	0.4016	0.7180	0.6022	0.6687

The quantitative comparison between the base model and the LoRA adapted model shows significant gains on all evaluation metrics. The fully trainable base model has validated loss of 1.4907, ROUGE-1 of 0.3880, ROUGE-2 of 0.2361 and ROUGE-L of 0.3244. On the other hand, the LoRA modified model improved the validation loss to 0.4016 and ROUGE-1 to 0.7180, ROUGE-2 to 0.6022, and ROUGE-L to 0.6687.

These results show that LoRA greatly increases summarization quality by simply altering a tiny fraction of model parameters. The significant improvement of ROUGE-2 is particularly noteworthy since it suggests better phrase-level consistency and better semantic alignment between generated and reference summaries.

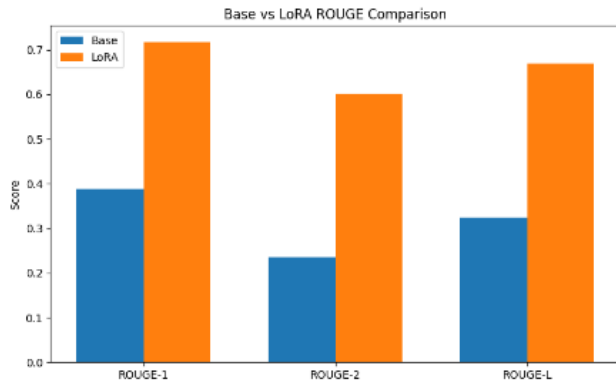


Fig1: Base vs LoRA ROUGE Comparison

Visualization of the ROUGE measures further demonstrates the superiority of the LoRA-adapted model. The training loss curve shows stable optimization and satisfactory convergence, however there are fluctuations due to the relatively small size of the dataset. Although this dataset is synthetic and organized, the findings clearly highlight the efficiency of parameter-efficient fine-tuning for scientific summarization.

Qualitative study of the output generated further validates the quantitative findings. The original model tended to generate brief, generic summaries whereas the LoRA-adapted model generated more thorough, structured and useful summaries that was a better representation of the input scientific information.

6. DISCUSSION

Experimental results establish that parameter-efficient fine-tuning is highly effective in scientific summarization tasks, especially in low-resource settings. LoRA efficiently fine-tunes for summarization by updating a small subset of parameters, keeping the pretrained information of the original model.

A significant observation is that the LoRA-adapted model performed better than the fully trainable model despite updating only 2.25% of total parameters. This demonstrates the ability of low-rank adaptation to focus the learning capacity on the task-relevant components of the transformer. Furthermore, the reduced number of trainable parameters greatly decreases the computing requirements, memory use and training cost.

The performance advantages we notice in this study are in part due to the structured and synthetic nature of the dataset. The model can learn summarization mappings more efficiently than it would on fully natural scientific corpora, since the data set follows relatively regular summary patterns. As such, the claimed ROUGE scores may be higher than actual performance on datasets like arXiv or PubMed.

7. LIMITATIONS AND FUTURE WORK

While the results are satisfactory, there are a few limitations. First of all, the data used in this work is synthetic and somewhat small. Real-world scientific data sets are much more linguistically variable and complex. Second, ROUGE measures don't fully reflect factual consistency or semantic accuracy. Third, the transformer has a sequence length limit, which forces us to truncate big scientific texts.

In future work, we will evaluate our approach on real-world scientific datasets such as arXiv and PubMed. We will also add more factual consistency criteria, such as QAFactEval, and human evaluation studies. Future work could explore larger instruction-tuned models and other parameter-efficient fine-tuning techniques.

8. CONCLUSION

In this research, it proposed an end-to-end scientific summarization system with FLAN-T5-small and LoRA-based parameter-efficient fine-tuning. Experiments show that LoRA considerably enhances the performance of summarization with a large reduction in trainable parameters and processing costs. The revised model generated high gains across ROUGE metrics and significantly reduced validation loss while updating only 2.25% of total parameters.

The results suggest parameter-efficient fine-tuning is a viable and successful approach for scientific NLP applications, especially in low-resource computing contexts.

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